A Framework to Analyze the Effects of PV Array Positioning and DC Sizing on Inverter Mission Profiles, Lifetime, and Reliability

Abstract– The inverter loading ratio (ILR) is traditionally selected to maximize energy yield, often overlooking converter lifetime. Compensating for sub-optimal PV array position—such as tilt and orientation—by over-sizing DC capacity acts as a hidden stress multiplier. This geometric compensation structurally reshapes the inverter's mission profile, exacerbating saturation duty-cycles and thermal stress. This paper proposes a comprehensive seven-phase lifetime-oriented sizing framework utilizing high-resolution field measurements. By parametrically scaling DC power across various array geometries, the framework evaluates the critical trade-off between marginal energy gains and converter degradation. Initial results from an equatorial baseline reveal that high DC over-sizing to offset geometric losses severely prolongs inverter exposure to high-stress clipping zones. This framework provides designers with a multidisciplinary tool to optimize PV design constrained by long-term physical reliability.

I. Introduction

In conventional practice, photovoltaic systems are increasingly designed under economic pressure to maximize energy yield while minimizing system cost. Consequently, ILR selection is driven primarily by energy-yield optimization and economic indicators such as the levelized cost of energy (LCOE) or payback time. To achieve these targets, system designers often adjust the installed DC capacity to compensate for sub-optimal panel positions, such as varying tilt angles and orientations.

However, inverter clipping introduces a strongly nonlinear relationship between incident irradiance and delivered AC power. When the available DC power exceeds the inverter rated capacity, output saturation truncates the upper tail of the power distribution. Because different array geometries directly reshape the plane-of-array irradiance, they alter the system's clipping losses and the temporal distribution of these saturation events. As a result, standard energy estimations may overlook how orientation and tilt influence the intensity of inverter clipping. [1]–[3].

Beyond energy losses, both panel position and ILR directly influence the inverter operating regime. Increasing ILR increases time spent near rated power and alters the frequency and duration of saturation events. Reliability-oriented design of power electronic converters relies on mission-profile-based analysis, where lifetime depends on the statistical distribution of operating conditions rather than solely on annual energy throughput. Consequently, systems delivering similar yearly energy—but with different array sizing and geometries—may experience substantially different stress exposure. [4]–[6].

While previous work has examined ILR-dependent lifetime variation, and others have quantified the impact of panel orientation and tilt on energy yield, these aspects have largely been treated independently. An integrated framework linking panel position, ILR sizing, mission profile extraction, and reliability-oriented indicators remains limited. This paper addresses this gap by proposing a comprehensive lifetime-oriented framework based on long-term field measurements of a grid-connected PV inverter. By combining parametric ILR scaling with different panel positions, the proposed approach enables the consistent evaluation of generated energy, clipping losses, mission profiles, inverter lifetime, and reliability.

II. Proposed Framework

The experimental baseline consists of a 5 kW single-phase grid-connected PV inverter with 5.13 kWp installed DC capacity, yielding a nominal ILR of approximately 1.03, sized to generate 750 kWh per month. The system installed in the Brazilian northeast has been continuously monitored from January 2019 to July 2025. Electrical quantities on both DC and AC sides, as well as meteorological variables, were recorded using calibrated sensors and a Campbell Scientific CR1000 datalogger. Data were acquired at high internal sampling rate and stored as 1-minute averages.

To systematically evaluate the impact of panel positioning and inverter loading ratio (ILR) on converter reliability, the data from this experimental baseline feeds into a comprehensive seven-phase framework as shown in Fig. 1.

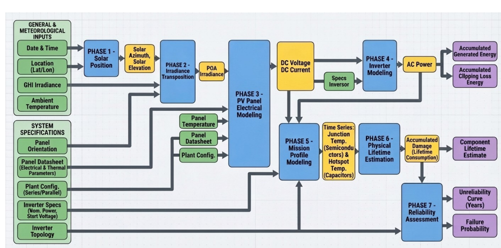

Figure 1: Diagram of Proposed Framework

The methodology links position and array sizing to electro-thermal stress and lifetime degradation. The partial results presented in this paper focus on the electrical and geometric outcomes up to Phase 4, while the full framework encompasses the complete reliability assessment.

Phase 1) Solar Position Calculation: The framework begins by determining the exact solar geometry over time using the installation's latitude and longitude .This phase calculates the solar elevation angle and the solar azimuth angle. These parameters govern the vectorial projection of direct irradiance onto any tilted surface.

Phase 2) Plane of Array (POA) Irradiance Calculation: This phase calculates the total irradiance reaching the panels. The global horizontal irradiance is first decomposed into direct and diffuse components using the Erbs model. Subsequently, it is transposed to the tilted surface using the Liu & Jordan isotropic model, yielding the POA irradiance.

Phase 3) Photovoltaic Panels Electrical Modeling: This phase translates the environmental conditions into electrical potential using the Single Diode Model. The I-V curve is solved analytically using the Lambert W function. The installed DC capacity is parametrically scaled to meet specific energy generation targets.

Phase 4) Inverter Modeling: The AC output power (P_{AC}) is reconstructed through an inverter model, which includes a finite state machine for startup conditions and a mathematically adjusted empirical efficiency curve (η).

Phase 5) Mission Profile Modeling: Advancing to the physical stress domain, this phase maps the electrical operation to component-level thermal responses. The inputs for this phase consist of the DC voltage and DC current, the reconstructed AC power, and the ambient temperature. These inputs are fed into a Foster RC thermal equivalent network to calculate power losses and generate the outputs, which are the continuous time-series profiles of the semiconductor junction temperatures (T_j) and capacitor hot-spot temperatures.

Phase 6) Physical Lifetime Estimation: This phase evaluates the physical degradation of the internal components. The junction and hot-spot temperatures are processed using a Rainflow counting algorithm to identify discrete thermal cycles. The outputs of this phase are the physical lifetime estimations and the accumulated damage of the components.

Phase 7) Reliability Evaluation: The final phase aggregates the component-level degradation to assess the system-level reliability. The output yields the unreliability curve over the years for the plant. This curve is modeled using a Weibull cumulative distribution function. This formulation provides the B_{10} metric, indicating the specific timeframe in which there is a 10% probability of system failure.

III. Experimental Results

For evaluation, a 16-day continuous dataset from March 2021 was analyzed. This period features high irradiance and a solar declination close to the zenith, providing a good stress-test scenario for the inverter.

To evaluate the impact of oversizing, the installed DC capacity was parametrically scaled. Fig. 2 illustrates the trade-off between the energy yield and the inverter's clipping behavior as the ILR increases. High oversizing yields marginal energetic gains but severely prolongs the converter's exposure to high-stress saturation zones.

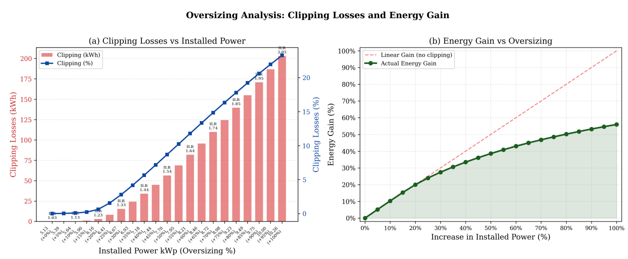

Figure 2: Effect of Oversizing

Fig. 3 evaluates the required installed DC power to meet a fixed generation target across four cardinal orientations and three tilt angles (5°, 10°, and 15°). Array orientation and tilt directly dictates the severity of the mission profile. Higher tilts present greater losses, requiring a higher installed capacity to achieve the same energy target, which effectively forces a higher ILR.

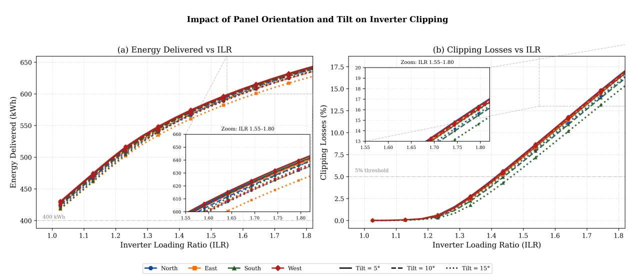

Figure 3: Effect of Orientation and Tilt

IV. Discussion, Conclusion and Future Work

The first four phases of the proposed framework were selected for presentation due to their critical relevance in establishing a reliable, foundation for the subsequent reliability analyzes. Furthermore, the results chosen for this preliminary version, derived from a 16-day evaluation window, demonstrate that compensating for non-optimal tilt angles (e.g. 15°) requires higher installed DC capacities to meet fixed energy generation targets. This position-driven over-sizing acts as a hidden stress multiplier. It structurally reshapes the inverter's mission profile by extending the saturation duty-cycle and exacerbating clipping losses.

The final version of this work presents a comprehensive set of results that generalizes the framework's application to any geographical location and varying temporal resolutions. Investigating these diverse scenarios is critical, as different latitudes impose distinct seasonal shifts in solar position, structurally altering geometric advantages, while lower sampling resolutions may obscure high-frequency irradiance peaks that trigger inverter saturation. By extending the analysis to diverse datasets, the study shows the tool's effectiveness in capturing seasonal shifts at different latitudes and high-frequency irradiance peaks. This approach will provide system designers with a multi-disciplinary tool to optimize ILR sizing constrained by the converter's long-term physical reliability.

References

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